Study Report: SQL for Data Analysis - Data Analyst

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Family:	Data Analyst
Level:	Intermediate
Chapters:	7
Source:	Reference: SQL for Data Analysis - Cathy Tanimura

Official sources and free libraries

- https://www.advanced-sql.com/

Summary

Study report for *SQL for Data Analysis* by Cathy Tanimura (Intermediate level) mapped to the Data Analyst role. Advanced SQL patterns for analytics in PostgreSQL and warehouses.

Chapters

Track: Book-Intro

Chapter 1: About SQL for Data Analysis [Intermediate]

Chapter focus: About SQL for Data Analysis *(Level: Intermediate)**

This study report summarizes *SQL for Data Analysis* by Cathy Tanimura for the Data Analyst role. The resource is rated Intermediate level. Advanced SQL patterns for analytics in PostgreSQL and warehouses. Use this report alongside the official material to prepare for CodeQuest review.

Code Reference:

```
# SQL for Data Analysis
# Author: Cathy Tanimura
# Role: Data Analyst
# Level: Intermediate
```

What it shows: Metadata block records the source book and target role family.

Topic guide

What it actually is
A structured study report based on SQL for Data Analysis.

When to use it
When learning Data Analyst skills at Intermediate level.

Where to use it
Data Analyst training paths and certification prep.

Real use example
A learner reads this report before starting the full book or free online guide.

Track: Book-Context

Chapter 2: Why This Book Matters for Your Role [Intermediate]

Chapter focus: Why This Book Matters for Your Role *(Level: Intermediate)**

Data Analyst professionals use ideas from SQL for Data Analysis to solve real workplace problems. Advanced SQL patterns for analytics in PostgreSQL and warehouses. This chapter explains how the book fits into your learning path and what you should be able to do after studying it.

Code Reference:

```
Role: Data Analyst
Book focus: Advanced SQL patterns for analytics in PostgreSQL and warehouses.
Recommended level: Intermediate
```

What it shows: Connects the book topic to the job role outcomes.

Topic guide

What it actually is
Role-aligned learning connects theory to job tasks.

When to use it
During career planning and syllabus design.

Where to use it
Data Analyst bootcamps and CodeQuest teacher assignments.

Real use example
A teacher assigns this book report before a intermediate skills checkpoint.

Track: Book-Topics

Chapter 3: Key Topics Covered [Intermediate]

Chapter focus: Key Topics Covered *(Level: Intermediate)**

The main topics in SQL for Data Analysis include practical concepts described as: Advanced SQL patterns for analytics in PostgreSQL and warehouses. Study each topic with hands-on practice. Take notes on definitions, workflows, and examples that match tools used in Data Analyst jobs today.

Code Reference:

```
- Topic 1: core concept
- Topic 2: core concept
- Topic 3: core concept
- Topic 4: core concept
```

What it shows: Topic list guides what to study chapter-by-chapter in the source.

Topic guide

What it actually is
Topic maps turn a book into actionable learning objectives.

When to use it
Before reading and while building chapter summaries.

Where to use it
Self-study, flipped classroom, and revision.

Real use example
A student maps each book chapter to a CodeQuest report section.

Track: Book-Applied

Chapter 4: Applied: SQL JOINs and Multi-Table Logic [Intermediate]

Chapter focus: SQL JOINs and Multi-Table Logic *(Level: Intermediate)**

JOINs combine facts with dimensions. INNER keeps matches only; LEFT keeps all left rows. Qualify ambiguous columns (orders.customer_id). Watch fan-out: joining orders to line-items multiplies rows - aggregate before join when needed.

Code Reference:

```
SELECT c.country,
       SUM(o.amount) AS revenue
FROM orders o
JOIN customers c ON c.id = o.customer_id
WHERE o.status = 'paid'
GROUP BY c.country;
```

What it shows: JOIN links orders to customers so revenue rolls up by country.

Topic guide

What it actually is
JOINs merge related tables on key columns.

When to use it
Any analysis spanning users, orders, products, or campaigns.

Where to use it
CRM exports, e-commerce warehouses, and subscription billing.

Real use example
Support tickets joined to account tier show churn risk concentrates in 'free' tier accounts.

Chapter 5: Applied: CTEs and Window Functions [Intermediate]

Chapter focus: CTEs and Window Functions *(Level: Intermediate)**

Common Table Expressions (WITH) break complex SQL into readable steps. Window functions (ROW_NUMBER, LAG, SUM OVER) compute metrics without collapsing rows - essential for funnels, running totals, and period-over-period growth.

Code Reference:

```
WITH daily AS (
  SELECT order_date, SUM(amount) AS revenue
  FROM orders GROUP BY 1
)
SELECT order_date, revenue,
       revenue - LAG(revenue) OVER (ORDER BY order_date) AS day_change
FROM daily;
```

What it shows: LAG compares each day to the previous - spot spikes without a self-join.

Topic guide

What it actually is
CTEs name subqueries; window functions aggregate over ordered partitions.

When to use it
Funnels, retention, rankings, and moving averages in SQL.

Where to use it
BigQuery, Snowflake, PostgreSQL, and DuckDB analytics.

Real use example
ROW_NUMBER deduplicates latest subscription per user before computing MRR.

Chapter 6: Applied: pandas groupby, merge, and pivot [Intermediate]

Chapter focus: pandas groupby, merge, and pivot *(Level: Intermediate)

groupby.agg computes segment metrics; merge joins DataFrames on keys (validate cardinality); pivot_table reshapes for heatmaps. Use categorical dtypes for faster groupby on large data.

Code Reference:

```
summary = (df.groupby('region')['amount']
             .agg(orders='count', revenue='sum', aov='mean')
             .sort_values('revenue', ascending=False))
merged = orders.merge(customers, on='customer_id', how='left')
```

What it shows: Named aggregations keep output columns clear for downstream charts.

Topic guide

What it actually is
pandas transforms mirror SQL but excel at notebook iteration.

When to use it
Between SQL export and visualization in a Python workflow.

Where to use it
Survey cross-tabs, SKU profitability, and multi-source blends.

Real use example
merge flags orphan orders (no matching customer) for data quality follow-up.

Track: Book-Plan

Chapter 7: Study Plan and Practice [Intermediate]

Chapter focus: Study Plan and Practice *(Level: Intermediate)

Finish SQL for Data Analysis with a weekly plan: read one section, write a summary, complete one exercise, and reflect on how it applies to Data Analyst. If a free edition exists, practice every example. Submit your notes and one mini-project demo for teacher review.

Code Reference:

```
Week 1: Read + notes
Week 2: Exercises
Week 3: Mini project
Week 4: Review + quiz
```

What it shows: A 4-week plan turns reading into demonstrable skill.

Topic guide

What it actually is
Structured study plans improve retention and portfolio outcomes.

When to use it
After finishing the key topics chapters.

Where to use it
CodeQuest end-of-unit assessments.

Real use example
A learner completes the study plan and uploads a intermediate project screenshot.